RL Homework 2 (Unit 5)

Self-Driving Lab: Reinforcement Learning for Experiment Design

Saniya Kousar

Computer Science Engineering(BE)

Roll No: 100523733092

Self-Driving Lab: RL for Experiment Design

RL Homework 2 (Unit 5) - Project 7 | 1. Motivation

- Discovering new materials means testing thousands of recipes; each experiment costs time, money and reagents
- Scientists change one setting at a time and rely on intuition - slow and easy to trap in a poor local optimum
- RL can learn which temperature and concentration to try next, using fewer experiments
- Goal: maximise reaction yield within a 25-experiment budget while staying safe

95%

maximum possible yield in our simulated reaction (80 C, 0.6 mol/L)

2. Method: Q-learning in a simulated lab

State

Current recipe: temperature (10 levels) x concentration (10 levels) = 100 states

Actions

Raise / lower temperature by 10 C, raise / lower concentration by 0.1 M

Reward

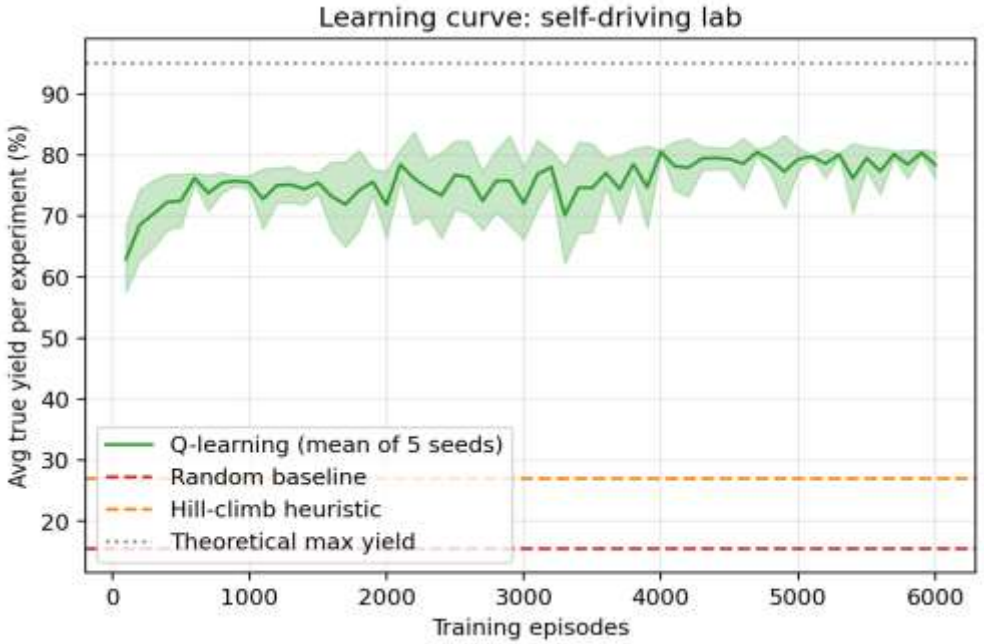
Measured yield - experiment cost - 0.3 penalty if T >= 110 C (unsafe)

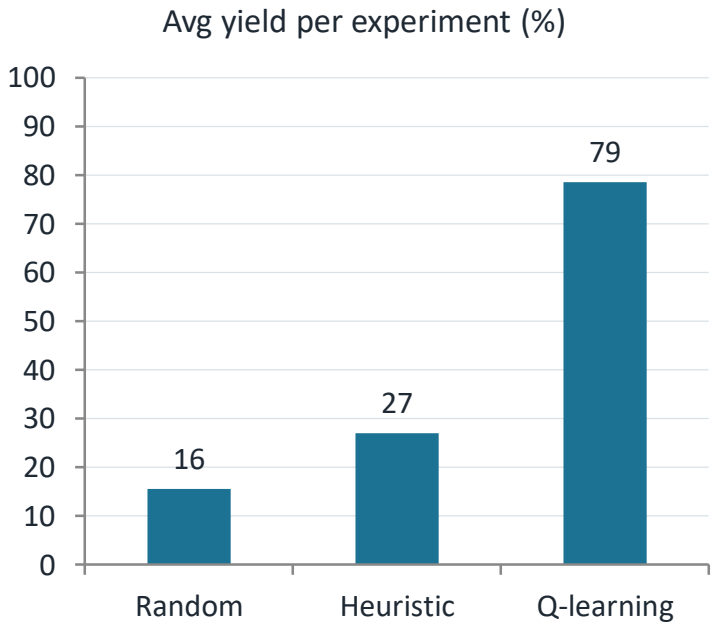
Algorithm

Tabular Q-learning, epsilon-greedy, alpha 0.05, gamma 0.9, 6000 episodes x 25 experiments

Environment: noisy yield surface with a decoy local optimum (45 C, 0.9 M) and decomposition above 100 C. Baselines: random and hill-climbing heuristic.

3. Results: RL beats both baselines





Unsafe experiments: random 20.1%, heuristic 11.5%, Q-learning 0.4%. Q-learning reaches the 95% optimum in every episode (best yield found).

4. Limitations

- Simulated reaction with a hand-made yield function - real chemistry has pH, time, catalyst and mixing effects
- Tiny discrete grid: tabular Q-learning cannot scale to many continuous variables (needs DQN, PPO or Bayesian optimisation)
- Policy is tied to one reaction; a new reaction needs retraining
- Time-independent policy sometimes oscillates between two good recipes (80 C <-> 70 C) near the end of an episode
- Training needs thousands of simulated episodes - impossible on a real lab without a simulator

5. Ethics and safety

Physical safety

Exploring blindly can cause runaway reactions. Use simulation first, hard safety limits and human sign-off.

Accountability

Scientists stay responsible for results. Log every experiment so findings are reproducible.

Jobs and access

Treat automation as an assistant to technicians; share tools so only rich labs do not benefit.

Conclusion: RL can speed up discovery, but safety and human oversight must come first.